

MINUSINSK, Russian Federation

WMO: 298660

Lat: 53.717N

Lon: 91.700E

Elev: 254

StdP: 98.30

Time Zone: 7.00 (E07)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.9	-35.5	-40.0	0.1	-37.7	-37.5	0.1	-35.4	8.4	-6.2	7.5	-7.0	0.0	110	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	31.0	19.2	29.2	18.4	27.4	17.8	20.5	28.2	19.6	26.5	18.8	25.1	2.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.4	22.6	17.4	12.8	22.2	16.5	12.1	21.8	60.4	28.3	57.4	26.5	54.8	25.1	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.1	6.1	-38.3	34.2	3.4	1.8	-40.7	35.5	-42.7	36.5	-44.6	37.5	-47.0	38.8		
(5)			-38.3	22.4	3.3	1.4	-40.7	23.4	-42.7	24.3	-44.6	25.1	-47.0	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-19.8	-16.3	-4.4	5.6	10.3	18.1	19.6	17.5	10.3	3.0	-6.6	-14.4	(6)
(7)		DBStd	14.69	8.71	8.40	7.22	4.77	4.55	3.73	2.70	3.12	3.99	4.81	7.43	8.31	(7)
(8)		HDD10.0	3827	926	736	447	145	55	1	0	0	43	221	498	756	(8)
(9)		HDD18.3	6089	1184	969	704	382	252	48	17	53	241	476	748	1014	(9)
(10)		CDD10.0	909	0	0	0	13	63	244	299	234	53	4	0	0	(10)
(11)		CDD18.3	130	0	0	0	0	2	42	57	28	1	0	0	0	(11)
(12)		CDH23.3	1741	0	0	0	7	89	628	612	357	48	0	0	0	(12)
(13)		CDH26.7	530	0	0	0	1	19	211	184	102	14	0	0	0	(13)
(14)	Wind	WSAvg	2.1	1.3	1.6	2.3	2.7	2.8	2.1	1.9	1.8	2.0	2.3	2.5	2.1	(14)
(15)	Precipitation	PrecAvg	355	8	7	6	16	35	57	67	63	45	26	14	12	(15)
(16)		PrecMax	532	32	30	25	38	95	180	137	183	119	61	48	30	(16)
(17)		PrecMin	222	0	0	0	2	3	12	20	17	5	1	0	2	(17)
(18)		PrecStd	61	6	5	5	9	17	28	28	35	23	12	9	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	3.5	14.7	24.2	29.0	33.4	33.1	32.3	28.6	19.1	10.0	2.8	(19)
(20)			MCWB	-1.2	0.6	6.3	11.8	15.7	19.6	20.9	20.2	16.7	11.3	5.7	-0.7	(20)
(21)		2%	DB	-0.9	1.5	10.8	20.8	25.8	31.0	30.6	29.4	23.8	15.6	7.4	0.9	(21)
(22)			MCWB	-2.8	-0.7	4.7	10.8	13.8	18.7	19.5	19.6	15.8	10.0	3.8	-1.4	(22)
(23)		5%	DB	-3.3	-0.4	8.0	18.0	23.0	29.2	28.6	27.3	20.8	13.1	4.9	-1.0	(23)
(24)			MCWB	-5.0	-2.1	3.3	9.3	12.6	17.9	18.9	18.6	14.1	8.3	2.1	-2.9	(24)
(25)		10%	DB	-6.6	-3.2	5.5	15.1	20.0	27.2	26.8	25.0	18.2	10.7	2.9	-2.6	(25)
(26)			MCWB	-8.0	-4.9	1.6	7.9	11.6	17.3	18.3	17.6	12.8	6.6	0.6	-4.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	1.1	6.8	13.0	16.7	21.4	21.9	21.5	18.2	12.0	6.3	0.6	(27)
(28)			MCDWB	0.7	3.1	13.4	23.4	26.4	30.0	29.0	29.2	26.7	17.4	9.4	2.5	(28)
(29)		2%	WB	-2.8	-0.5	5.0	11.2	15.0	20.1	20.5	20.2	16.3	10.3	4.1	-1.5	(29)
(30)			MCDWB	-0.9	1.2	10.3	19.4	23.4	28.0	28.1	27.1	22.2	15.1	6.7	0.5	(30)
(31)		5%	WB	-4.9	-2.4	3.5	9.9	13.7	19.1	19.7	19.1	15.0	8.7	2.3	-2.8	(31)
(32)			MCDWB	-3.3	-0.3	7.8	16.9	21.1	26.4	26.4	25.3	20.2	12.6	4.6	-0.9	(32)
(33)		10%	WB	-8.0	-5.1	1.8	8.3	12.3	18.1	19.0	18.3	13.5	7.0	0.6	-4.2	(33)
(34)			MCDWB	-6.7	-3.3	5.3	14.6	19.0	25.0	25.0	23.8	17.1	10.2	2.9	-2.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.1	13.2	12.7	13.2	13.3	13.7	11.6	12.2	12.7	10.4	8.1	9.2	(35)
(36)			MCDBR	10.2	11.6	15.0	19.4	20.1	18.5	16.3	16.7	18.5	15.8	8.8	8.0	(36)
(37)			MCWBR	8.9	9.4	9.4	10.0	9.5	7.3	6.5	7.4	10.3	9.9	6.1	6.6	(37)
(38)		5% WB	MCDWB	10.0	11.3	14.4	18.0	16.5	15.2	13.5	14.4	16.6	15.7	8.6	7.7	(38)
(39)			MCWBR	8.8	9.2	9.1	9.5	8.2	6.5	5.8	6.7	9.3	10.1	6.2	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.319	0.368	0.409	0.416	0.405	0.417	0.403	0.365	0.348	0.323	0.289	(40)	
(41)		taud	2.052	2.005	1.977	2.134	2.176	2.311	2.299	2.309	2.351	2.269	2.111	2.002	(41)	
(42)		Ebn at Noon	642	742	787	814	831	845	826	816	802	718	586	537	(42)	
(43)		Edh at Noon	73	107	138	136	137	121	121	113	95	83	69	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.01	2.04	3.50	4.46	5.28	6.08	5.50	4.61	3.23	1.79	0.90	0.65	(44)	
(45)		RadStd	0.07	0.14	0.28	0.25	0.51	0.53	0.36	0.36	0.31	0.16	0.10	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page